

Research-level Problem:

[Q6] For a graph $G = (V, E)$, let $G_S = (V, E(S, S))$ denote the graph with the same vertex set, but only the edges between vertices in S . Let L_S be the Laplacian matrix of G and let L_S be the Laplacian of G_S . I say that a set of vertices S is ϵ -light if the matrix $\epsilon L - L_S$ is positive semidefinite. Does there exist a constant $c > 0$ so that for every graph G and every ϵ between 0 and 1, V contains an ϵ -light subset S of size at least $c\epsilon|V|$?

Mathematicians:

Yes, the constant is $1/42$.

Lemma B.1 (Final Bound Lemma): $\exists S$ with $|S| \geq \epsilon n/42$ and $\epsilon L \geq L_S$.

Claim B.2 (Leverage Score Property): For S and T subsets of vertices, $l(S, T) = \text{Tr}(\tilde{L}_{S, T})$.

Lemma B.3 ($\ell\Phi$ -control): $\exists t \notin S, s.t. \Phi_\sigma^{u+\delta}(S \cup \{t\}) \leq \phi$ and $\ell(S \cup \{t\}) \leq \ell(S) + 4$.

Lemma B.4(ℓ -control): $>$ half of $t \notin S$ have $\ell(S \cup \{t\}) \leq \ell(S) + 4$.

Claim B.5: $\forall T \subset V, \sum_{t \in T} \ell(t, T) \leq 2(|T| - 1)$.

Lemma B.7 (Φ -control): If $\frac{\text{Tr}_\sigma(M^{-2}B)}{\Phi_\sigma^u(A) - \Phi_\sigma^{u+\delta}(A)} + \text{Tr}_\sigma(M^{-1}B) < 1$, then $\Phi_\sigma^{u+\delta}(A + B) \leq \Phi_\sigma^u(A)$.

Lemma B.8 (Upper Barrier): $\geq$ half of $t \notin S$ have $U(S, t) < 1$. $\Rightarrow$ The barrier doesn't increase.

Lemma B.9 (Barrier for Π_S): $\sum_{t \notin S} \text{Tr}_\sigma \left((u + \delta)\Pi_S - \tilde{L}_S \right)^p \tilde{L}_{S, T} \leq \text{Tr}_\sigma(M^p)$.

Aletheia (DeepMind, 2026):

Aletheia has found **no solutions** within the time limit or output **no solutions found**.

GPT-5.2R (OpenAI, 2026):

Yes, the constant is $1/256$. [Looser Bound]

Step 1: Normalization on the Laplacian range. [Correct]

Step 2: A one-sided BSS barrier lemma: $\text{tr}(BU) + \frac{\text{tr}(BU^2)}{\Phi_u(M) - \Phi_{u'}(M)} \leq 1$. [Correct]

Step 3: A partial coloring process, color k vertices one by one using r colors. [Correct]

Step 4: Inductive barrier invariant: $M_{t+1} < u_{t+1}I$ and $\Phi_{u_{t+1}}(M_{t+1}) \leq \Phi_{u_t}(M_t)$. [Hallucinated References]

Step 5-6: Extracting a Larger ϵ -light set and size lower bound. [Looser Bound]

Ours:

Yes, the constant is $1/42$. [Correct]

Theorem 1 (Final Bound Lemma): $\exists S$ with $|S| \geq \epsilon n/42$ and $\epsilon L \geq L_S$. [Correct]

Claim 2 (Reduction): it suffices to prove Theorem 1 in connected graphs. [Correct]

Lemma 3 (Trace Formula for Leverage Scores): $\sum_{e \in E} l_e^M = \text{Tr}(M^\dagger M)$. [Correct]

Lemma 4 (Quadratic): S is ϵ -light if and only if $\sum_{\{u, v\} \in E(S, S)} (f_u - f_v)^2 \leq \epsilon \sum_{\{u, v\} \in E} (f_u - f_v)^2$. [Correct]

Lemma 5 (Greedy Step): S is ϵ -light if and only if $\sigma_M(v) = \sum_{u \in S, \{v, u\} \in E} l_{\{v, u\}}^M \leq 1$. [Correct]

Lemma 6 (Good Vertex): there is a good vertex that we can add to the set. [Correct]

Lemma 7 (Induction): Finish induction with the greedy algorithm and make conclusion. [Correct]